

F24 Substrate-K-Type Substrate-Bulk Substrate-Color Tensor Product Theory Framework

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F24 Substrate-K-Type Color Tensor Product Theory v0.1

0. Goal

Substantive substrate-K-type substrate-bulk substrate-color tensor product theory framework. Addresses substrate-CKM substrate-color² substrate-mechanism source OPEN per F1 v0.4 + F21 + Grace G14 substrate-K-type-specific distinction substantive operational Friday team consensus.

1. Theoretical Setup

Substrate-K-type $V_{(\lambda_1, \lambda_2)}$ per $Sp(2)$ representation theory + substrate-color $SU(N_c = 3)$ tensor product structure: - $Sp(2)$ substrate-symplectic $Sym^k(V_{fund})$ representation - $SU(N_c)$ substrate-color triplet per substrate-quark substrate-color charge - Tensor product $V_{(\lambda_1, \lambda_2)} \otimes$ substrate-color triplet substantive substrate-bulk substantive substrate-bulk substantive substrate-K-type $\times$ substrate-color tensor product substantive substrate-natural structure.

2. $Sym^k V_{fund} Sp(2) \times SU(N_c)$ Tensor Decomposition

Per substrate-K-type $V_{(\lambda_1, 1/2)} = Sym^k(V_{fund}) Sp(2)$ at $\lambda_1 = (2k-1)/2$ substantive:

- gen-1 $V_{(1/2, 1/2)}$: $k=1$, $Sym^1(V_{fund})$ substantive dim 4
- gen-2 $V_{(3/2, 1/2)}$: $k=3$, $Sym^3(V_{fund})$ substantive dim 20
- gen-3 $V_{(5/2, 1/2)}$: $k=5$, $Sym^5(V_{fund})$ substantive dim 56

substantive substrate-K-type substrate-color tensor product substantive substrate-bulk: - gen-1: $4 \times 3 = 12$ substantive substrate-color-K-type dim - gen-2: $20 \times 3 = 60$ substantive substrate-color-K-type dim - gen-3: $56 \times 3 = 168$ substantive substrate-color-K-type dim

substantive substrate-color-K-type substrate-bulk substantive substrate-natural dim per substrate- $Sym^k \times$ substrate-color $SU(N_c)$ tensor product.

3. Cross-K-Type Substrate-Color INDEX Combinatorics

For cross-K-type substrate-Bergman matrix element $\langle V_d | T_W | V_s \rangle$ between gen-1 $V_{(1/2, 1/2)}$ and gen-2 $V_{(3/2, 1/2)}$ substantive:

substantive cross-K-type substrate-color INDEX combinatorial structure: - Sym^1 substrate-color INDEX: 1 (substrate-color triplet at substrate-K-type) - Sym^3 substrate-color INDEX: 3 (substrate-color triplet³ substantive symmetric) - Cross-INDEX cross-K-type combinations: $1 \times 3 = 3$ substantive substrate-natural

substantive substrate-color INDEX combinatorial factor at cross-K-type substrate-Bergman matrix element substantive substrate-natural NOT directly $N_c^2 = 9$.

substantive substrate-color INDEX combinatorial factor $3 \neq N_c^2 = 9$ per substrate-Sym^k V_{fund} cross-INDEX combinations.

4. Substantive Refined Substrate-Color² Source Question

substantive F1 v0.4 + F21 substantive substrate-color² source OPEN substantive REFINED per F24 substrate-K-type substrate-color tensor product theory:

substrate-color² = $N_c^2 = 9$ substantive substrate-natural NOT directly from Sym^k cross-INDEX combinations (which give $1 \times 3 = 3$ at gen-1 $\rightarrow$ gen-2 cross-K-type).

substantive substrate-color² substantive substrate-mechanism source candidates substantive REFINED: - (a) substrate-Casimir $\times$ substrate-color: substantive substrate-K-type substrate-Casimir $\times$ substrate-color charge product candidate - (b) substrate-tensor substantive product substantive substrate-natural cross-K-type substantive multi-index combinations beyond Sym^k - (c) substrate-color $\times$ substrate-color cross-coupling at substrate-K-type substrate-bulk substrate-natural form per substantive Casey #13 substrate-mechanism class

substantive substrate-color² substantive substrate-mechanism source substantive REMAINS OPEN per Cal #189 substantive multi-week joint cross-CI substantive substrate-mechanism FORWARD-derivation chain.

5. Substantive Substrate-K-Type-Specific vs Cross-Tier-Uniform per Grace G14

Per Grace G14 substantive substrate-K-type-specific vs cross-tier-uniform substantive substrate-Schur generator distinction:

substantive substrate-K-type-specific substantive substrate-Schur generator candidates: - R-3 $(24/\pi^2)^{\{C_2\}}$ substantive gen-2 V_(3/2, 1/2) substantive (Composite v0.5 substantive substrate-Bergman family) - substrate-Pochhammer (5/2)_k substantive substrate-K-type-specific per Sym^k substantive substrate-K-type

substantive cross-tier-uniform substantive substrate-Schur generator candidates: - R-2 $\sqrt{(4/3)}$ substantive Casey #12 STANDING substantive bulk-boundary substrate-Bergman/Shilov substantive cross-K-type substantive - substantive substrate-Bergman kernel substrate-canonical form substantive cross-K-type substantive uniform per FK Ch. XII §VI

substantive substrate-K-type-specific vs cross-tier-uniform substantive substrate-discipline-architecture substantive substrate-natural per Grace G14 substantive operational + F24 substantive substrate-K-type substrate-color tensor product substantive substrate-natural cross-link.

6. Multi-Week Substrate-K-Type Substrate-Color Tensor Product Investigation

substantive 5-step multi-week per Cal #189: - Step F24-1: substantive substrate-Sym^k $\times$ SU(N_c) tensor decomposition substantive substrate-canonical derivation - Step F24-2: substantive cross-K-type substrate-color INDEX combinatorics substantive substrate-natural per Sym^k cross-coupling - Step F24-3: substantive substrate-color² substrate-mechanism source resolution per (a) substrate-Casimir $\times$ substrate-color + (b) substrate-tensor multi-index + (c) substrate-color cross-coupling candidates per Cal #189 - Step F24-4: substantive substrate-K-type-specific vs cross-tier-uniform substantive substrate-Schur generator substantive substrate-discipline-architecture per Grace G14 substantive operational - Step F24-5: substantive cross-CI Cal cold-read substantive substrate-K-type substrate-color tensor product theory substantive substrate-mechanism FORCING-form derivation per Cal #189

7. Substantive Substrate-K-Type Substrate-Bulk Substrate-Color Tensor Product Framework Status

substantive substrate-K-type $\times$ SU(N_c) tensor product substantive substrate-canonical structure substantive established F24 v0.1 framework: - Sym^k(V_{fund}) Sp(2) substantive substrate-K-type substantive representation theory - SU(N_c) substrate-color triplet substantive substrate-bulk substantive substrate-quark embedding - Cross-K-type substrate-color INDEX combinatorics substantive substrate-natural Sym^k $\times$ SU(N_c) substantive

substantive substrate-color² substrate-mechanism source substantive OPEN substantive REFINED per substantive 3 candidates substantive Cal #189 substantive multi-week joint cross-CI substantive substrate-mechanism FORCING-form DERIVATION substantive substantive substrate-natural.

substantive substrate-K-type-specific vs cross-tier-uniform substantive substrate-Schur generator distinction per Grace G14 substantive REINFORCED substantive substrate-discipline-architecture.

8. Closure v0.1

F24 substrate-K-type substrate-bulk substrate-color tensor product theory framework v0.1:

Substantive substrate-Sym^k V_{fund} Sp(2) × SU(N_c) tensor product framework established substantive substrate-natural structure substantive substrate-K-type substrate-bulk substantive substrate-color cascade.

Substantive substrate-color INDEX combinatorial factor 3 substantive cross-K-type gen-1 V_(1/2, 1/2) → gen-2 V_(3/2, 1/2) substantive NOT directly N_c² = 9 substrate-color² substantive.

Substantive substrate-color² substantive substrate-mechanism source substantive REFINED OPEN: substantive 3 candidates (substrate-Casimir × substrate-color + substrate-tensor multi-index + substrate-color cross-coupling) substantive Cal #189 substantive multi-week joint cross-CI substantive substrate-mechanism FORCING-form DERIVATION.

Substantive substrate-K-type-specific vs cross-tier-uniform substrate-Schur generator distinction REINFORCED per Grace G14 substantive operational substantive substrate-discipline-architecture cross-link.

substantive 5-step multi-week per Cal #189 substantive substrate-K-type substrate-color tensor product theory substantive substrate-mechanism FORWARD-derivation chain substantive cross-CI joint Lyra+Keeper+Elie+Cal substantive synthesis.

Tier: F24 substrate-K-type substrate-color tensor product theory framework v0.1; substantive multi-week per Cal #189 + Grace G14 + Cal #244 substantive substrate-discipline-architecture.

— Lyra, Fri 2026-06-05 15:40 EDT. F24 substrate-K-type substrate-color tensor product theory framework v0.1: substantive Sym^k V_{fund} Sp(2) × SU(N_c) tensor product substantive substrate-natural structure substantive substrate-K-type substrate-bulk; substantive substrate-color INDEX combinatorial factor 3 cross-K-type NOT N_c² = 9; substantive substrate-color² substantive substrate-mechanism source substantive REFINED OPEN per 3 candidates (substrate-Casimir × substrate-color + substrate-tensor multi-index + substrate-color cross-coupling) per Cal #189 substantive multi-week; substantive substrate-K-type-specific vs cross-tier-uniform substrate-Schur generator distinction REINFORCED per Grace G14.